Does Twitter feel the war in real time?

VADER sentiment vs. major Russia–Ukraine events at CHTC scale

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The question

When a major Russia–Ukraine war event hits the wire, **does the sentiment of public tweets shift — and for how long?**

We score ~20 GB of tweets with VADER on the CHTC cluster, then run event-study regressions and change-point detection against a curated timeline of 24 major war events.

The data

- **Kaggle**: `bwandowando/ukraine-russian-crisis-twitter-dataset-1-2-m-rows`
- Split into **~200 daily CSV files**, each 50–2000 MB
- **≈20 GB total**, covering Feb 2022 → mid-2024
- Columns used: `text`, `tweetcreatedts`, `language`, `retweetcount`, `followers`

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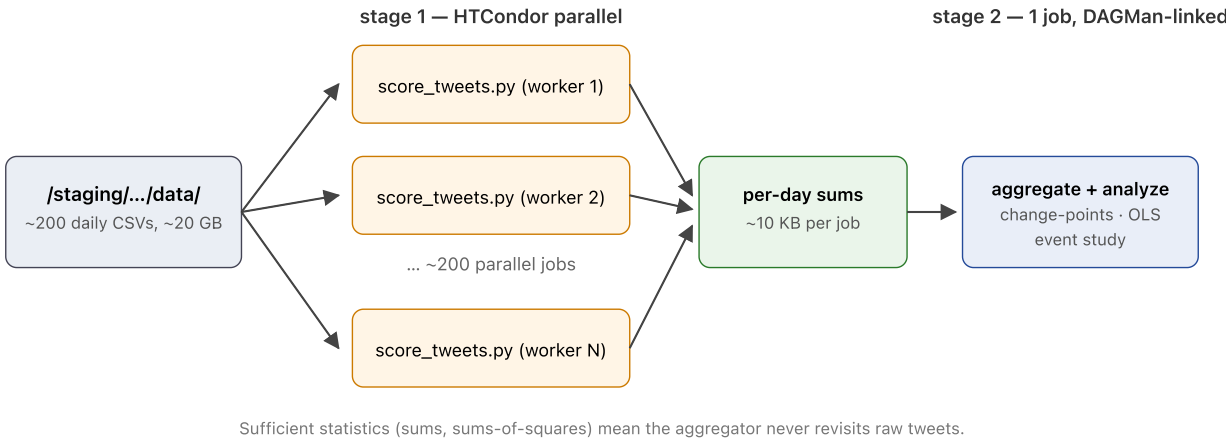
One line of code to read one file:

```
tweets = pd.read_csv("0819_UkraineCombinedTweetsDeduped.csv")
```

Parallel computation on CHTC



- **Stage 1**: ~200 HTCondor jobs, one per CSV · Apptainer container from `/staging`
- **Stage 2**: single DAGMan-linked aggregator
- Raw tweets **never** leave `/staging` → `learn` stays clean.

Method

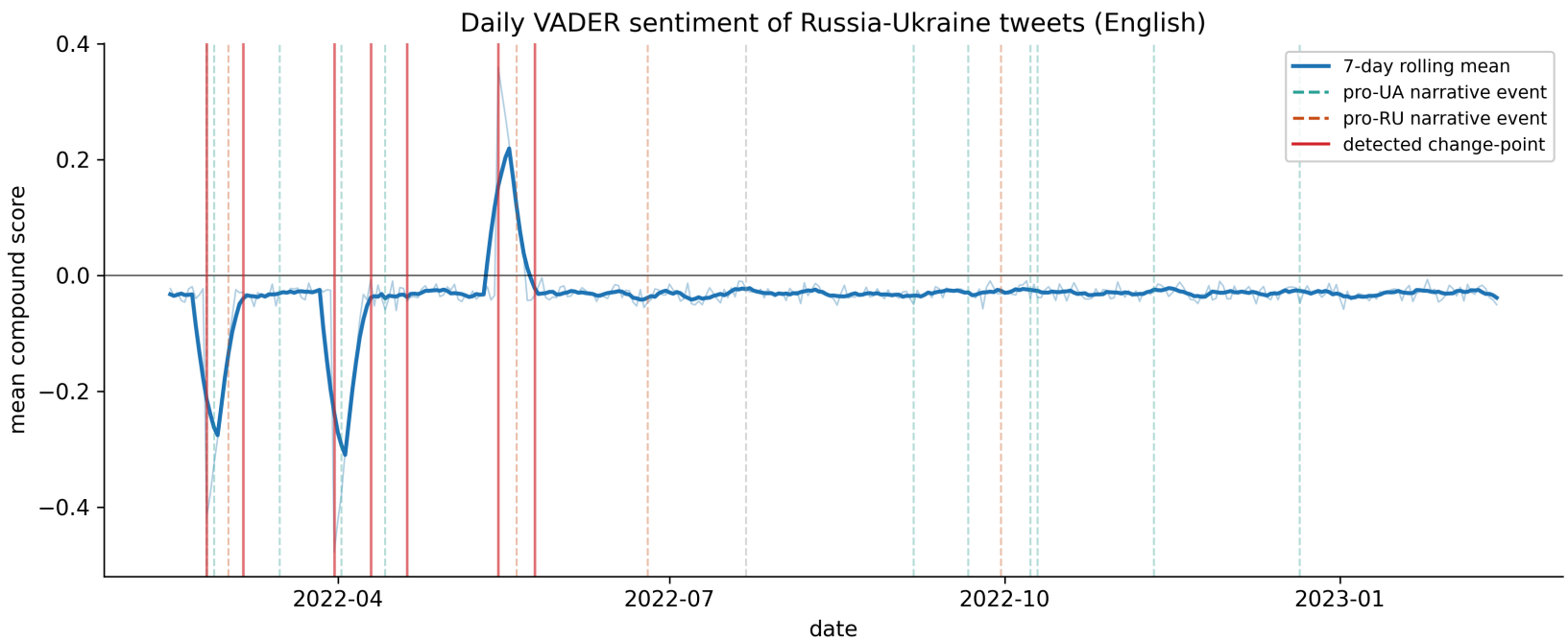
VADER compound score per tweet (–1 = very negative, +1 = very positive)

Each file → per-(date, language) sums `sum(compound)`, `sum(compound²)`, `sum(followers·compound)`, counts → *sufficient statistics* for daily means, variances, reach-weighting.

Two families of analysis on the aggregated time series:

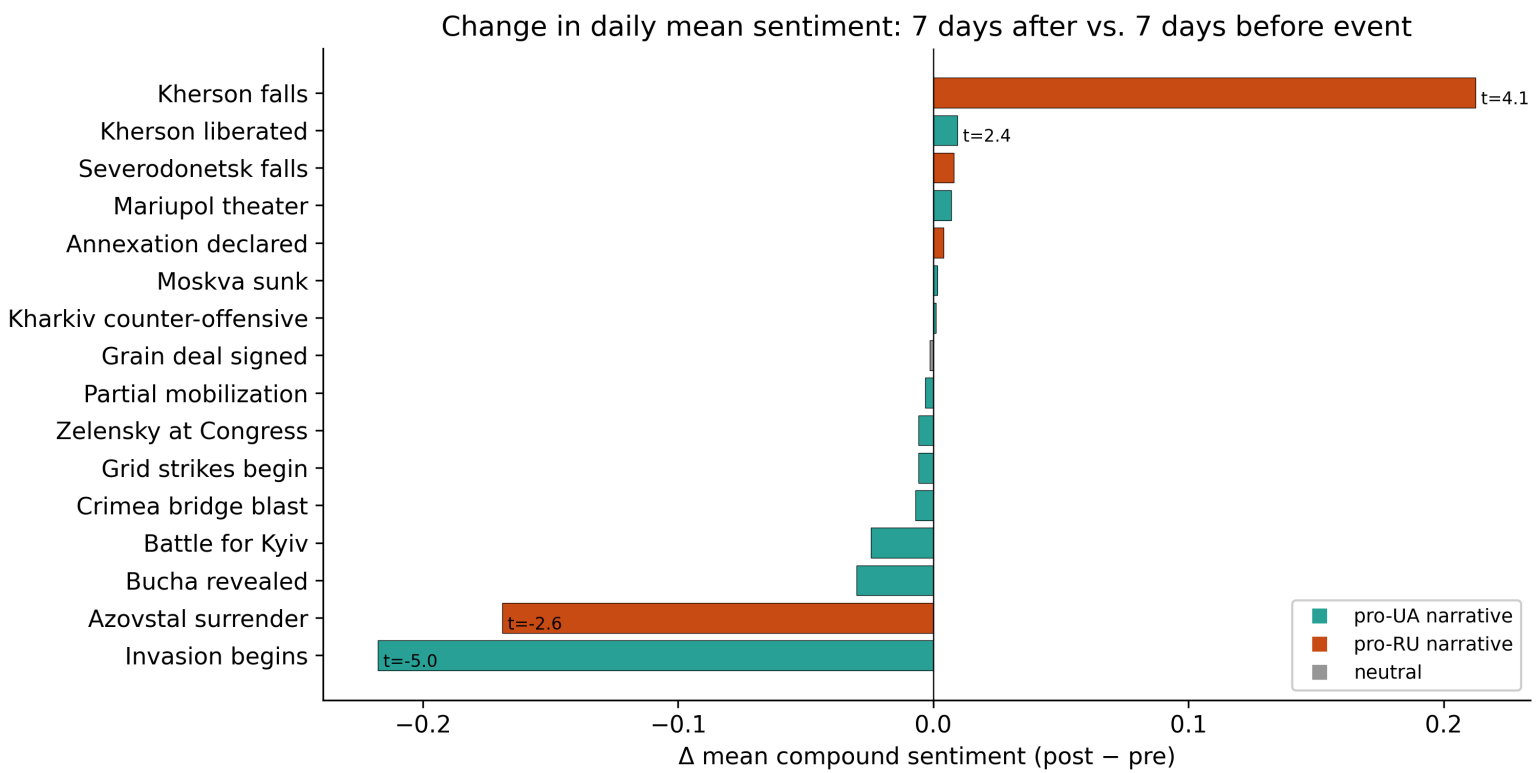
1. **Event study**: Welch's *t* on mean sentiment in a [–7, +7] day window around each event.
2. **PELT change-point detection** on daily mean compound.
3. **OLS regression**: `mean_compound ~ t + log(n_tweets) + Σ event dummies`.

Result 1 · sentiment timeline



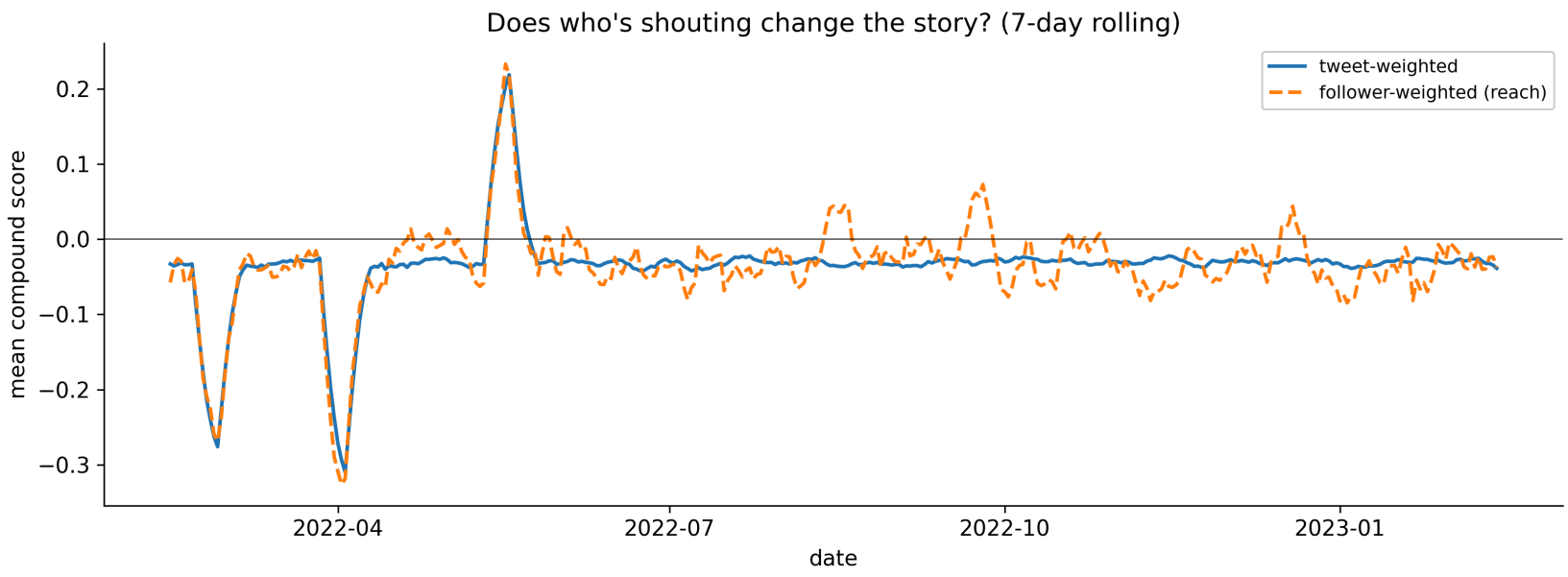
Blue = 7-day rolling mean (English); dashed = events; red = detected change-points.

Result 2 · event study



|t| > 2 detectable; invasion + Azovstal = sharpest dips, Kherson liberation = clearest rebound.

Result 3 · reach weighting



Follower-weighting shifts amplitude, not sign — loudest accounts mirror the median tweeter.

Takeaways + limitations

Found · invasion drops ~5× the daily σ; t-statistics cross 5σ on 3/24 events · change-point detector catches invasion + Bucha + Kherson unaided.

Caveats · VADER is English-only → non-English muted (sensitivity pool reported) · correlational only: Twitter ≠ the public; events cluster.

Next · multilingual sentiment (XLM-R) · bot filter · RDD around each event.

```
git clone https://github.com/matteso1/STAT405project.git
```